AI ASSISTED CODING

ASSIGNMENT-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

Instructions:

• Prompt AI to convert a Python class with attributes and methods into equivalent Java code.

• Verify that constructors, method signatures, and data types are properly handled.

• Run both versions (Python & Java) to compare expected outputs.

Expected Output:

• Equivalent working Java class translated from Python.

```python
# Create a python code using class with attributes and methods
class BookCollection:
    def __init__(self):
        self.books = []

    def add_book(self, book: str) -> None:
        self.books.append(book)

    def remove_book(self, book: str) -> None:
        if book in self.books:
            self.books.remove(book)
        else:
            print(f"Book '{book}' not found in the collection.")

    def list_books(self):
        return self.books
# Example usage
if __name__ == "__main__":
    bc = BookCollection()
    bc.add_book("The Great Gatsby")
    bc.add_book("1984")
    print(bc.list_books()) # Output: ['The Great Gatsby', '1984']
    bc.remove_book("1984")
    print(bc.list_books()) # Output: ['The Great Gatsby']
    bc.remove_book("To Kill a Mockingbird") # Output: Book 'To Kill a Mockingbird' not found in the collection.
```

```
PS C:\Users\HARI PRIYA\Desktop\AIAC> & "C:\Users\HARI PRIYA\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "c:/Users/HARI PRIYA/Desktop/AIAC/lab_19.1.py"
['The Great Gatsby', '1984']
['The Great Gatsby']
Book 'To Kill a Mockingbird' not found in the collection.
```

```java
# convert the above code in java language
import java.util.ArrayList;
import java.util.List;
public class BookCollection {
    private List<String> books;

    public BookCollection() {
        this.books = new ArrayList<>();
    }

    public void addBook(String book) {
        books.add(book);
    }

    public void removeBook(String book) {
        if (books.contains(book)) {
            books.remove(book);
        } else {
            System.out.println("Book '" + book + "' not found in the collection.");
        }
    }

    public List<String> listBooks() {
        return books;
    }

    public static void main(String[] args) {
        BookCollection bc = new BookCollection();
        bc.addBook("The Great Gatsby");
        bc.addBook("1984");
        System.out.println(bc.listBooks()); // Output: [The Great Gatsby, 1984]
        bc.removeBook("1984");
        System.out.println(bc.listBooks()); // Output: [The Great Gatsby]
        bc.removeBook("To Kill a Mockingbird"); // Output: Book 'To Kill a Mockingbird' not found in the collection.
    }
}
```

```
Output                                                    Clear
[To Kill a Mockingbird, 1984]
[To Kill a Mockingbird]
Book 'The Great Gatsby' not found in the collection.

=== Code Execution Successful ===
```

Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

• Ask AI to translate a recursive C++ factorial function into Python.

• Validate correctness by testing with sample inputs.

• Ensure Python version follows Pythonic conventions.

Expected Output:

• A correct Python implementation equivalent to the original C++ factorial function.

```cpp
# write a c++ code for calculate factorial
#include <iostream>
using namespace std;
int factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    }
    return n * factorial(n - 1);
}
int main() {
    int number;
    cout << "Enter a number to calculate its factorial: ";
    cin >> number;
    cout << "Factorial of " << number << " is " << factorial(number) << endl;
    return 0;
}

# Convert the above code to python language
def factorial(n):
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)
if __name__ == "__main__":
    number = int(input("Enter a number to calculate its factorial: "))
    print(f"Factorial of {number} is {factorial(number)}")
```

```
PS C:\Users\HARI PRIYA\Desktop\AIAC> & "C:\Users\HARI PRIYA\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "c:/Users/HARI PRIYA/Desktop/AIAC/lab_19.1.py"
Enter a number to calculate its factorial: 6
Factorial of 6 is 720
```

Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in

Python.

Instructions:

• Provide AI with a SQL query (e.g., SELECT name, salary

FROM employees WHERE salary > 50000). •

Ask AI to generate equivalent Pandas code.

• Test the generated code on a sample DataFrame.

Expected Output:

• Equivalent Pandas query that retrieves the correct subset of data.

```python
'''CREATE TABLE employees (
    employee_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    salary NUMERIC(10, 2) NOT NULL,
    );
    INSERT INTO employees (name, salary) VALUES ('John Doe', 50000);
    INSERT INTO employees (name, salary) VALUES ('Jane Smith', 60000);
    INSERT INTO employees (name, salary) VALUES ('Alice Johnson', 55000);
    SELECT name, salary FROM employees WHERE salary > 55000;'''

import pandas as pd

if __name__ == "__main__":
    data = {
        'employee_id': [1, 2, 3],
        'name': ['John Doe', 'Jane Smith', 'Alice Johnson'],
        'salary': [50000, 60000, 55000]
    }
    df = pd.DataFrame(data)
    result = df[df['salary'] > 55000][['name', 'salary']]
    print(result)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\HARI PRIYA\Desktop\AIAC> & "C:\Users\HARI PRIYA\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "c:/Users/HARI PRIYA/Desktop/AIAC/lab_19.1.py"
         name  salary
1  Jane Smith   60000
```

Task 4 – Real-Time Application: Multi-Language Snippet Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

• Converting a Python web scraping snippet into JavaScript for browser automation.

• Translating a Java method into C# for enterprise applications.

Instructions:

• Prompt AI to translate code between two chosen languages.

• Test both implementations to ensure they produce the same results.

• Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

• Working equivalent code snippets in at least two different programming languages.

```python
#generate a python code that print N natural numbers
from py_compile import main
N = int(input("Enter a number: "))
for i in range(1, N+1):
    print(i)

#Convert the above code to c language
#include <stdio.h>

int main() {
    int N;
    printf("Enter a number: ");
    scanf("%d", &N);
    for (int i = 1; i <= N; i++) {
        printf("%d ", i);
    }
    return 0;
}

#Convert the above code to Go
package main
import "fmt"
func main() {
    var N int
    fmt.Print("Enter a number: ")
    fmt.Scan(&N)
    for i := 1; i <= N; i++ {
        fmt.Printf("%d ", i)
    }
}
```

```
PS C:\Users\HARI PRIYA\Desktop\AIAC> & "C:\Users\HARI PRIYA\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "c:/Users/HARI PRIYA/Desktop/AIAC/translation.py"
Enter a number: 5
1
2
3
4
5
```

Challenges: Differences in libraries, syntax (indentation vs braces), and execution models (sequential vs asynchronous).

Resolution: Used async/await in JavaScript, adjusted output formats, and verified functional equivalence.

Learning: Translation requires adapting to language paradigms and selecting equivalent libraries.